Exercise Sheet

Lecture 6

Diagonalization, Cayley–Hamilton, Functions of Operators

Throughout, the core uses finite-dimensional V over the stated $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $T \in \mathcal{L}(V)$, same-field functions $f : \mathrm{spec}(T) \to \mathbb{F}$, and $\chi_T(\lambda) = \det(\lambda I - T)$ is the characteristic polynomial. In the quantum preview, $|v\rangle$ is simply Dirac notation for the vector v; bras are introduced in Lecture 7. Tags mark the flavour; a $\star$ marks a harder one.

Core tutorial route

Work **6.1, 6.2, 6.7, 6.11, 6.14, 6.15, 6.17, and 6.26** in that order (about 80–100 minutes before discussion). This eight-problem route requires a complete diagonalization, a defective diagnosis, powers, an authentic Cayley–Hamilton reduction and proof, a finite matrix exponential, dynamical transfer, and spectral mapping.

Additional practice: 6.3–6.6, 6.8–6.10, 6.12–6.13, 6.16, 6.21–6.22. **Bridge:** 6.19, 6.23–6.24 (convergence and matrix dynamics). **Extension:** 6.18, 6.20, 6.25, 6.27–6.28 (minimal-polynomial recognition, Jordan/defective exponentials, ladder operators, and a stronger determinant identity). Extension tasks do not count toward core progress.

Diagonalization

Exercise 6.1 (computational) Diagonalize $A = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}$: find P and D with $P^{-1}AP = D$.

Exercise 6.2 (computational) Decide whether $A = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$ is diagonalizable, with reasons.

Exercise 6.3 (computational) For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (eigenvalues $3, 1$, eigenvectors $(1,1), (1,-1)$) compute A^n for general n.

Exercise 6.4 (computational) Diagonalize $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ over $\mathbb{R}$: find P and D with $P^{-1}AP = D$.

Exercise 6.5 (computational) Diagonalize $A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$ over $\mathbb{C}$.

Exercise 6.6 (computational) Decide whether $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ is diagonalizable, with reasons.

Exercise 6.7 (computational) For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ compute A^n for general $n \geq 0$.

Exercise 6.8 (proof) Suppose $T \in \mathcal{L}(V)$ is diagonalizable and has only one eigenvalue λ. Prove that $T = \lambda I$.

Cayley–Hamilton

Exercise 6.9 (computational) Use the Cayley–Hamilton theorem to express A^{-1} as a polynomial in A for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Exercise 6.10 (computational) Use the Cayley–Hamilton theorem to express A^{-1} as a polynomial in A for $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$.

Exercise 6.11 (computational) For $A = \left(\begin{smallmatrix} 0 & 1 \\ -2 & 3 \end{smallmatrix}\right)$, use Cayley–Hamilton to write A^4 in the form $\alpha A + \beta I$, and give the resulting matrix.

Exercise 6.12 (computational) For $A = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{smallmatrix}\right)$ compute χ_A and use Cayley–Hamilton to express A^{-1} as a polynomial in A; give A^{-1} explicitly.

Exercise 6.13 (proof) Suppose $T \in \mathcal{L}(V)$ is invertible on a complex space of dimension n. Prove that T^{-1} is a polynomial in T.

Exercise 6.14 (proof) Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. By direct computation show that $A^2 - (\operatorname{tr} A)\, A + (\det A)\, I = 0$, thereby verifying Cayley–Hamilton for every 2×2 matrix.

Functions of operators

Exercise 6.15 (computational) For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ compute e^{tA}.

Exercise 6.16 (proof) For an arbitrary $T \in \mathcal{L}(V)$, prove directly that for any polynomials p, q, the operators $p(T)$ and $q(T)$ commute. Do not assume diagonalizability.

Exercise 6.17 (computational) Let $A = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)$. Compute e^{tA} and describe the solution curves of $\dot{x} = Ax$.

Exercise 6.18 ($\star$) **Extension (minimal-polynomial recognition).** Suppose T is diagonalizable with eigenvalues in $\{0, 1\}$ (a projection-like operator). Prove that $T^2 = T$.

Exercise 6.19 ($\star$) Suppose A is diagonalizable and every eigenvalue of A satisfies $|\lambda| < 1$. Prove that $A^n \to 0$ as $n \to \infty$ (entrywise).

Exercise 6.20 (computational) **Extension (Jordan–Chevalley and defective exponential).** Write $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)$ as $A = D + N$ with D diagonalizable, N nilpotent, $DN = ND$, and use it to compute e^{tA}.

Exercise 6.21 (computational) For $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ compute e^{tA}.

Exercise 6.22 (computational) Find the positive square root $\sqrt{A}$ of $A = \left(\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right)$ (the one with positive eigenvalues), so that $(\sqrt{A})^2 = A$.

Exercise 6.23 (computational) Compute e^{tA} for $A = \left(\begin{smallmatrix} 0 & 3 \\ -3 & 0 \end{smallmatrix}\right)$.

Exercise 6.24 (computational) Solve $\dot{x} = Ax$ with $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ and $x(0) = (2, 0)$.

Exercise 6.25 (computational) **Extension (defective exponential).** Compute e^{tA} for $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{smallmatrix}\right)$.

Exercise 6.26 (computational) Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, with $\operatorname{spec}(A) = \{1, 3\}$. Using spectral mapping, find $\operatorname{spec}(B)$ for $B = A^3 - 2A + I$.

Exercise 6.27 ($\star$) **Extension (finite ladder analogy).** On the spin-1 representation the raising operator acts by $J_+|{-1}\rangle = \sqrt{2}\,|0\rangle$, $J_+|0\rangle = \sqrt{2}\,|1\rangle$, $J_+|1\rangle = 0$. Write its matrix in the basis $(|1\rangle, |0\rangle, |{-1}\rangle)$ and show J_+ is nilpotent.

Exercise 6.28 (proof) **Extension.** Let A be diagonalizable. Prove that $\det(e^A) = e^{\operatorname{tr} A}$.